

ANA FAHIM | Hidden Radiology

QUALITY & SAFETY MASTERY

Vol. 4: Quality Assurance, Radiation Protection,
Contrast Safety & Patient Risk Management

Hidden Radiology Educational Series
Author & Consultant: Nasser Mabrouk
Senior Radiology Specialist

See Beyond The Image



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CHAPTER 1: Radiation Protection Principles & ALARA Optimization

Radiation safety in diagnostic imaging focuses on minimizing radiation exposure to patients, staff, and the public while maintaining necessary diagnostic image quality. Protection protocols rely on established radiobiological safety principles.

1.1 Biological Effects of Ionizing Radiation

Stochastic Effects: Non-threshold, probabilistic effects (e.g., radiation-induced carcinogenesis or genetic mutations). The probability of occurrence increases with dose, but severity is independent of the absorbed dose.

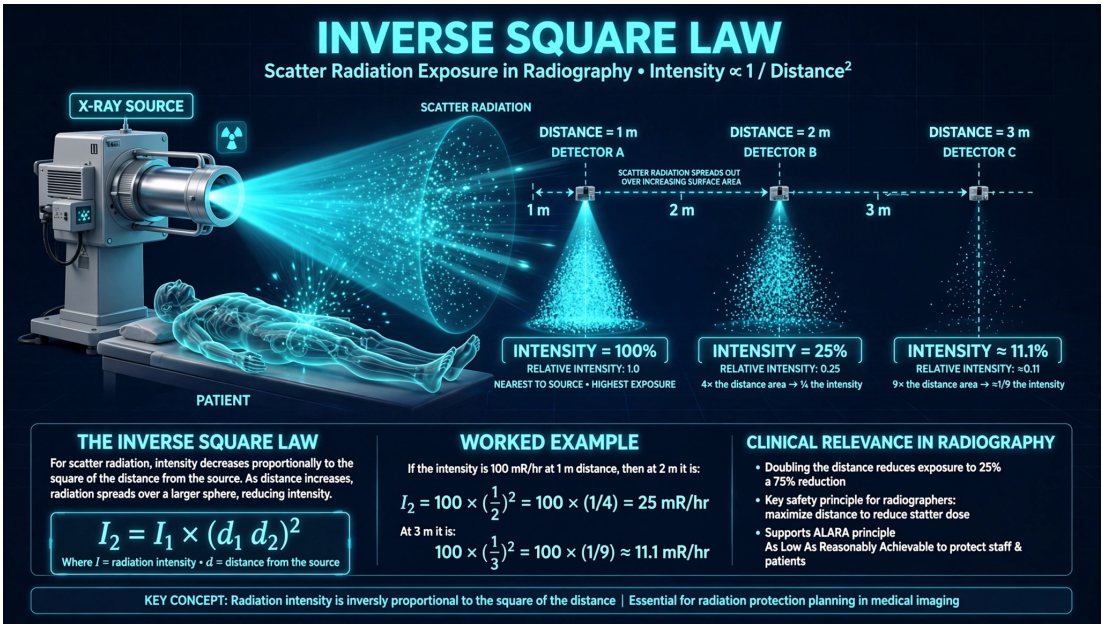
Deterministic (Tissue Reaction) Effects: Threshold-dependent effects occurring only after exceeding a specific radiation dose (e.g., skin erythema, epilation, cataracts). Severity increases directly with dose.

1.2 Fundamental Protection Framework (ALARA)

The ALARA principle (As Low As Reasonably Achievable) rests on three main protective factors:

Protection Factor	Physical Principle	Clinical Application
Time	Dose $\propto$ exposure duration.	Minimize fluoro beam-on time; pulsed fluoro.
Distance	Inverse Square Law: $I_1/I_2 = D_2^2/D_1^2$	Step back 2m during mobile X-ray or C-arm.
Shielding	Attenuation absorbs scatter.	Lead aprons 0.35-0.5 mm Pb, thyroid collar, glasses.

Illustration 1.1: Inverse Square Law - Doubling distance reduces dose to 25%.

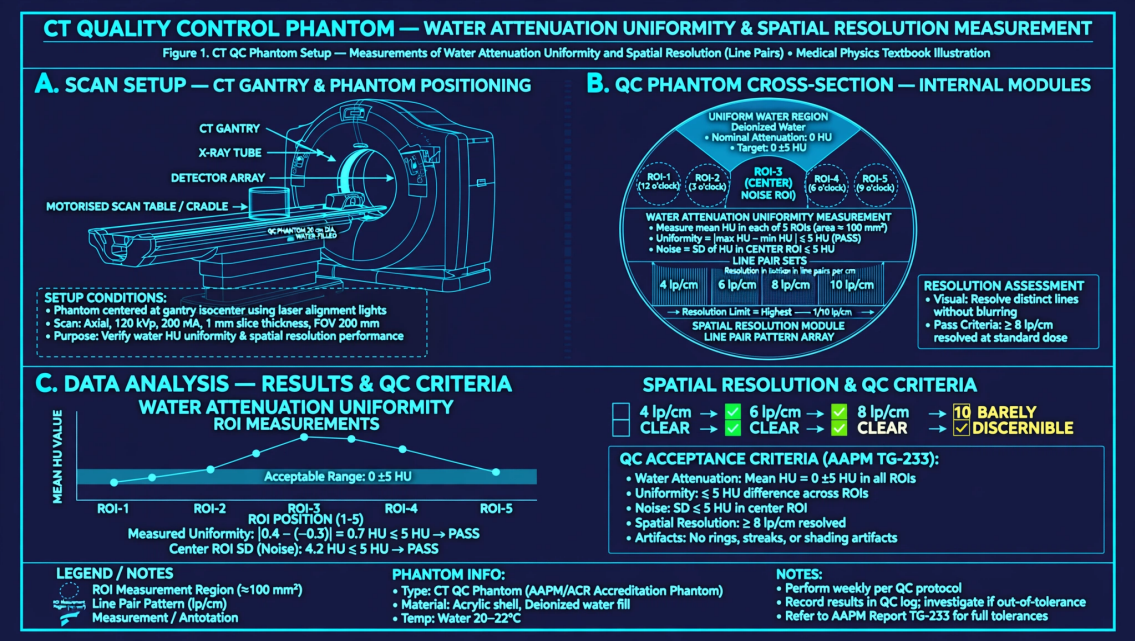


CHAPTER 2: Diagnostic Quality Control (QC) & Equipment Testing Protocols

Quality Control (QC) programs ensure that diagnostic imaging equipment operates within specific operational tolerances to guarantee optimal image quality at the lowest reasonable radiation dose.

QC Test	Frequency	Tolerance	Action
KVP Accuracy	Annual	±5% of set kVp	Recalibration >5%
Exposure Linearity	Annual	$L \leq 0.10$	Timer/tube adjustment
Collimator Alignment	Semi-Annual	$\leq 2\%$ of SID	Mirror/light adjustment
CT Water Calibration	Daily/Weekly	0 ± 3 HU	Recalibrate grid
CT Uniformity	Weekly	≤ 5 HU variability	Beam hardening check

Illustration 2.1: CT phantom calibration - water uniformity 0 HU & spatial resolution.



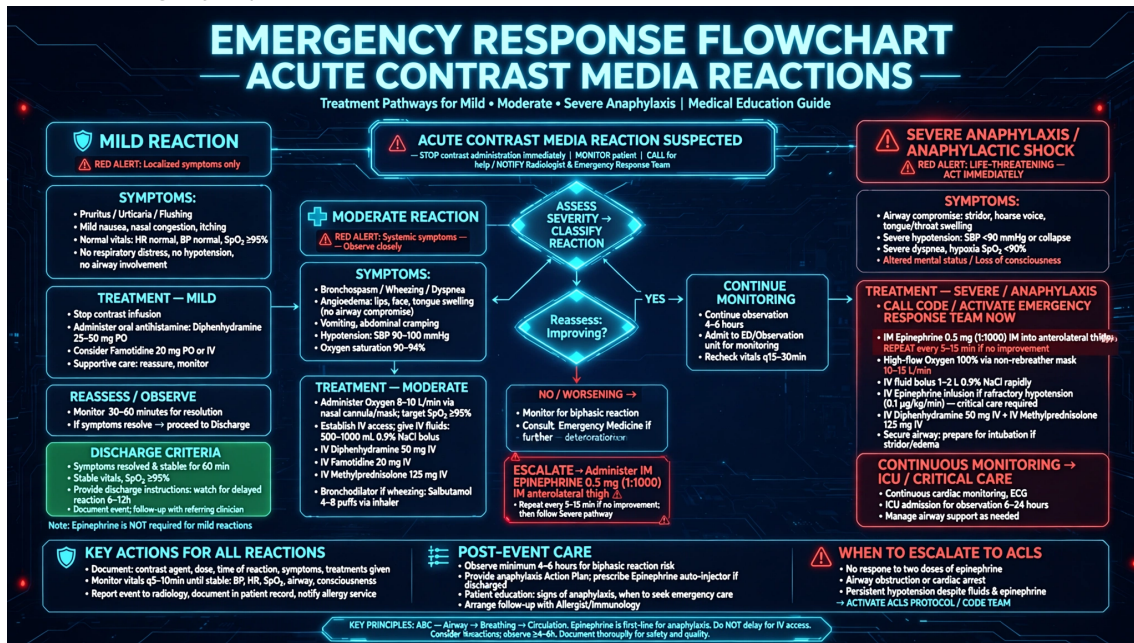
CHAPTER 3: Contrast Media Safety, Adverse Reactions & Emergency Response

IV contrast administration carries risks of hypersensitivity and PC-AKI. Technologists must be trained in early identification and emergency management.

Severity	Symptoms	Action
Mild	Nausea, limited urticaria, warmth, itching	Monitor, reassure, vitals
Moderate	Widespread urticaria, bronchospasm, edema, stridor, wheezing	Stop contrast, O2, Diphenhydramine, Beta-2 agonist
Severe	Severe bronchospasm, hypotension, anaphylaxis	Code Blue, airway, O2, Epinephrine IM 0.3mg, IV fluids

3.2 PC-AKI Prevention: eGFR ≥ 45 Safe; 30-44 Caution + hydration; <30 Avoid unless essential. Hydration: 0.9% NS 1-1.5 mL/kg/h 6-12h pre/post.

Illustration 3.1: Emergency response flowchart for acute contrast reactions.



CHAPTER 4: Infection Control & Ergonomics in Medical Imaging

Standard Precautions: WHO 5 Moments hand hygiene. Gloves, masks, eye protection during invasive procedures.

Decontamination: Disinfect transducers, sponges, plates between patients.

Airborne: N95 for TB, 12 air changes/hour in negative pressure.

4.2 Ergonomics

Use slide boards, lifts, multi-person transfers. Ultrasound: neutral wrist, avoid shoulder abduction $>30^\circ$, adjust monitor/couch heights.

CHAPTER 5: Patient Identification, Risk Management & Clinical Pre-Flight Checklists

Category	SOP / Requirement	Check
Patient ID	Verify two IDs (Name & DOB/MRN) vs wristband	☐ Pass
Procedure & Site	Confirm modality, site, indication	☐ Pass
Pregnancy	Screen females 12-55y before ionizing radiation	☐ Pass
Shielding	Lead apron integrity annual fluoro check	☐ Pass
Emergency Equipment	Crash cart, AED, O2, suction daily in contrast areas	☐ Pass
Incident Reporting	Document malfunctions, dose spikes, near-misses in QMS	☐ Pass

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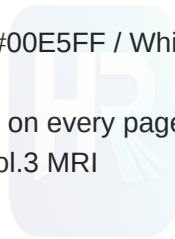
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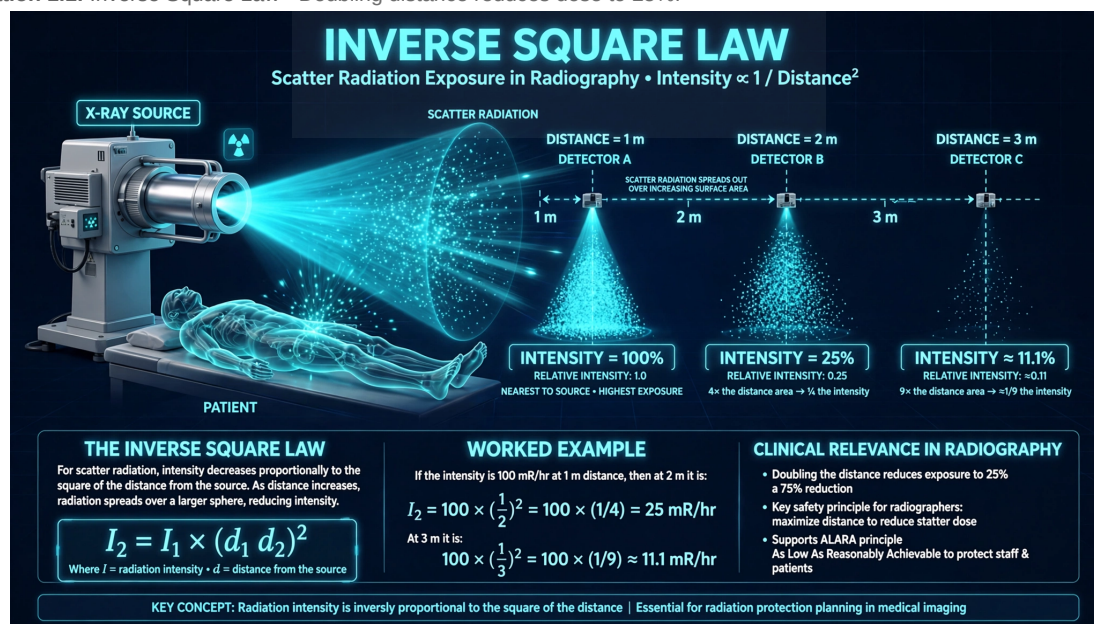
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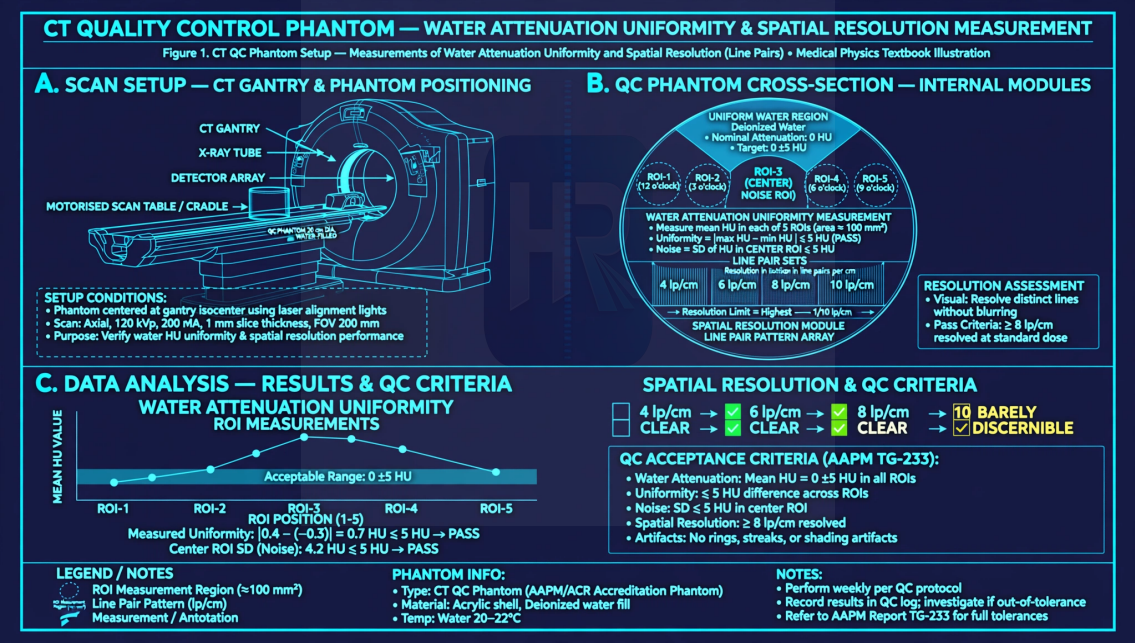


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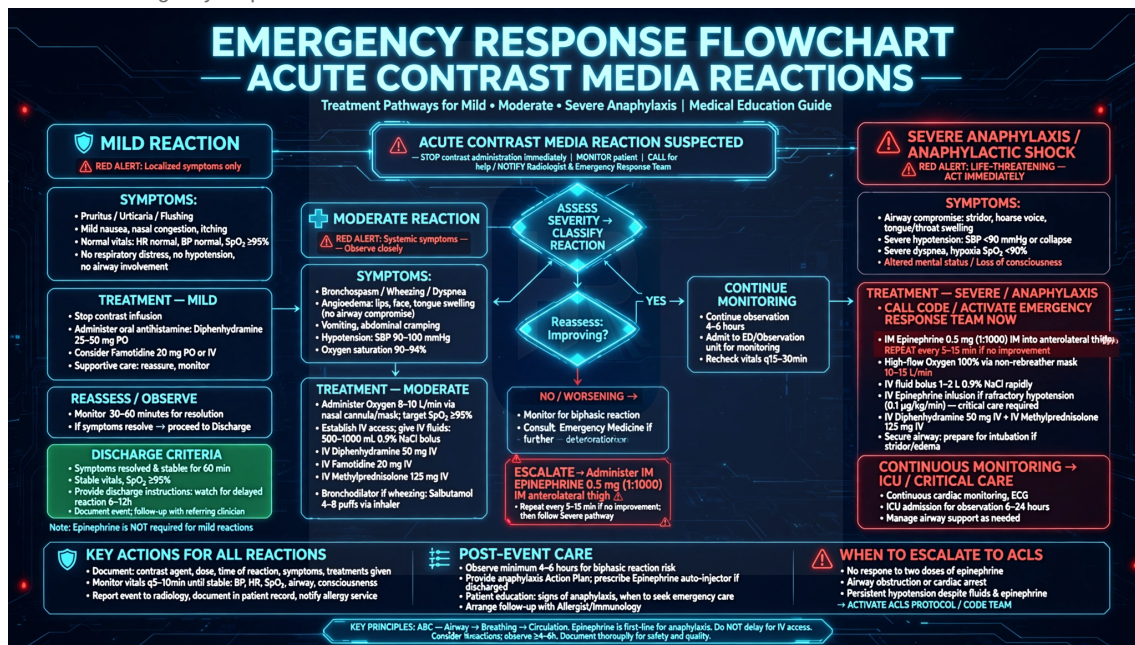
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